Leakage-Aware Brain Tumour MRI Classification with a Calibrated, Explainable EfficientNet-B0

A white paper on data hygiene, explanation quality and an in-browser research demo

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Abstract

Public brain MRI collections are widely used to benchmark four-way tumour classifiers (glioma, meningioma, pituitary tumour, no tumour), and reported accuracies often approach 99%. We revisit the popular Kaggle *Brain Tumor MRI Dataset* (version 2, 7,200 images) with an emphasis on evaluation hygiene. A perceptual-hash audit shows that, although no file is byte-identical across the official Training and Testing folders, 441 of the 1,600 Testing images (27.6%) have a strict near-duplicate in Training, concentrated in the no-tumour class. We remove every Training image linked to a Testing image, split the remainder into train and validation sets by duplicate group, and fine-tune an ImageNet-pretrained EfficientNet-B0 on a CPU. On the full Testing folder the model reaches 93.4% accuracy (95% bootstrap CI 92.1%–94.5%), macro-F1 0.933 and macro one-vs-rest ROC-AUC 0.989, while validation accuracy inside the training distribution is 98.6%. An identical model trained on the original folders scores 96.1%, but it is right on 105/105 Testing meningiomas that have near-copies in Training against 74/105 for the leakage-aware model, showing how much of the usual headline accuracy reflects memorisation. Temperature scaling ($T = 1.84$) reduces the expected calibration error on Testing from 0.054 to 0.036, and a confidence threshold chosen on validation flags 3.8% of Testing predictions as uncertain. We compare five attribution methods (Grad-CAM, Grad-CAM++, Integrated Gradients, occlusion and KernelSHAP on superpixels) with the model-parameter randomisation sanity check and deletion/insertion faithfulness metrics: Grad-CAM passes the sanity check and leads on insertion, whereas Integrated Gradients fails the sanity check despite the best deletion score. The model, exported to ONNX, runs in the browser with exact Grad-CAM, occlusion and a templated "why this prediction" panel that uses only model outputs and measured test statistics. The system is a research and education artefact, not a medical device.

1 Introduction

Brain tumours are a heterogeneous group of neoplasms whose diagnosis rests on imaging, histology and, since the 2021 WHO classification, molecular markers [13]. Magnetic resonance imaging is the main imaging modality, and automatic slice-level classification of MRI into broad tumour categories has become a standard exercise for transfer learning in medical imaging [4, 8, 24]. Much of this work uses small public collections in which several images come from the same patient or the same scan, and in which the provenance of individual images is only partly documented.

Two problems follow. First, when near-identical slices sit on both sides of a train/test split, test accuracy partly measures memorisation rather than generalisation, a form of leakage that has repeatedly inflated results in applied machine learning [11, 20]; even curated vision benchmarks contain such near-duplicates [5]. Second, the saliency maps that often accompany these classifiers are rarely tested: they can look plausible while being insensitive to the model's parameters [2], and in medical images they frequently fail to localise pathology [3, 21].

This white paper documents a small, fully reproducible project that tries to address both. Our contributions are:

1. an audit of exact and near-duplicate images in version 2 of the Kaggle Brain Tumor MRI Dataset [16], with all hashes, pairs and split files released for review;
2. a leakage-aware split and an EfficientNet-B0 [25] baseline trained on a CPU, reported with confidence intervals, per-class metrics, ROC-AUC, calibration and an explicit uncertainty threshold, and compared with a naive split of the same data;
3. a comparison of five attribution methods with sanity checks and faithfulness metrics; and
4. an in-browser demonstrator that reproduces the PyTorch probabilities, computes Grad-CAM exactly from ONNX outputs, and explains each prediction using only measured quantities.

2 Related work

Brain tumour classification. Cheng et al. [7] released the figshare T1-weighted contrast-enhanced collection (glioma, meningioma, pituitary tumour) and classified tumour regions with handcrafted features after region augmentation and partitioning. Deepak and Ameer [8] and Swati et al. [24] showed that features from ImageNet-pretrained networks [9], fine-tuned block-wise, reach high accuracy on the same data, and Badža and Barjaktarović [4] reported a compact CNN evaluated with patient-wise and record-wise cross-validation, noting the gap between the two. The Kaggle dataset used here merges figshare with the SARTAJ and Br35H collections [16]. Volumetric multi-sequence benchmarks such as BraTS [15] target segmentation of gliomas rather than slice-level type classification.

Leakage and duplicates. Kapoor and Narayanan [11] catalogue leakage as a main cause of irreproducible machine-learning findings, and Roberts et al. [20] found that most COVID-19 imaging models suffered from data duplication across sources and splits. Barz and Denzler [5] purged near-duplicates from CIFAR and measured the resulting accuracy drop. We use perceptual hashing [27] combined with pixel correlation to find such pairs.

Explanations and their evaluation. Occlusion sensitivity [28], Grad-CAM [22] and Grad-CAM++ [6], Integrated Gradients [23], LIME [19] and SHAP [14] are common post-hoc attribution methods. Adebayo et al. [2] proposed the model-parameter randomisation test, and Petsiuk et al. [18] introduced the deletion and insertion metrics. Benchmarks in radiology report that saliency maps often fall short of localisation targets [3, 21]. Calibration of modern networks and temperature scaling are due to Guo et al. [10].

3 Data and duplicate audit

3.1 Dataset

Version 2 of the Brain Tumor MRI Dataset (13 February 2026, CC BY 4.0) holds 7,200 2-D slices in a `Training` folder (1,400 per class) and a `Testing` folder (400 per class). According to its description, glioma, meningioma and pituitary images come from figshare [7] and SARTAJ, no-tumour images come from Br35H, and version 2 removed duplicates and eliminated train/test overlap. Images vary in size (most are 512×512) and in sequence and orientation (axial, coronal and sagittal; T1, T1 with contrast, T2 and FLAIR); no patient identifiers are provided. We download it with `kagglehub` and never redistribute it, apart from eight resized sample slices in the demo, as the licence allows with attribution.

3.2 Audit method

For each image we compute the MD5 of the file, a 64-bit DCT perceptual hash (pHash, `imagehash`) of the greyscale image and a 64×64 zero-padded thumbnail. For every pair within a Hamming

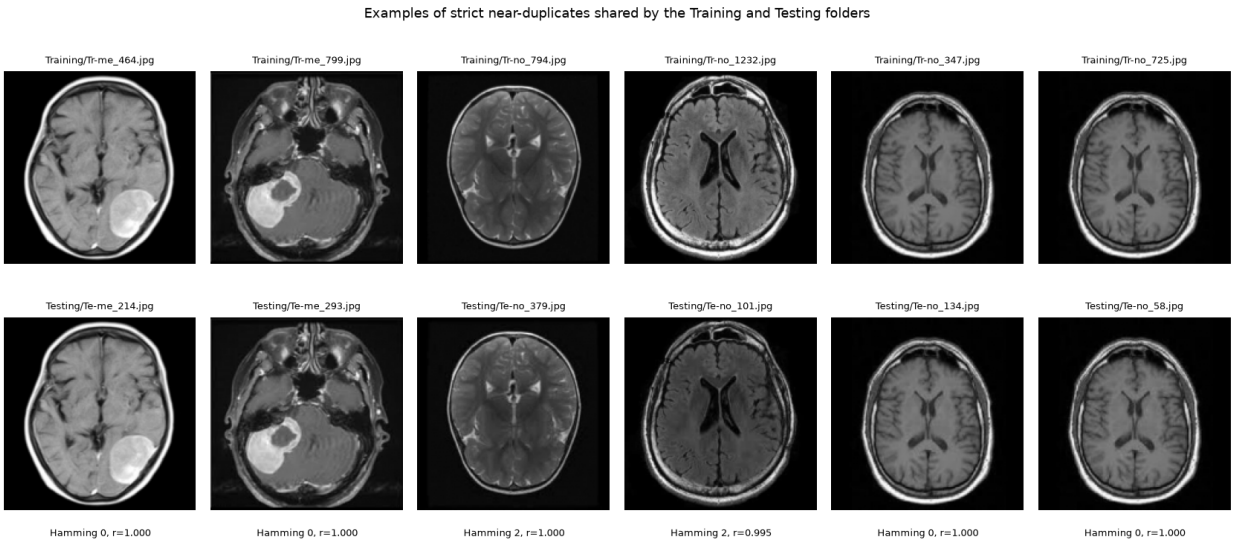


Figure 1: Strict near-duplicates shared by the Training and Testing folders (top: Training file, bottom: matching Testing file, with pHash Hamming distance and thumbnail correlation).

distance of 10 we compute the Pearson correlation of the thumbnails. Visual inspection of random pairs at each distance showed that pHash alone is too coarse for MRI (many different slices of similar anatomy sit 4–6 bits apart), so we define:

- **strict near-duplicate**: identical MD5, or Hamming distance ≤ 2 and correlation ≥ 0.98 (in our inspection, the same image re-encoded or resized);
- **linked**: identical MD5, or Hamming distance ≤ 6 and correlation ≥ 0.95 (also catches near-identical neighbouring slices of the same scan).

Connected components of linked images form *groups*. Thresholds were fixed by inspection before any model was trained; the full candidate-pair list (Hamming ≤ 10) is released so that others can apply different thresholds.

3.3 Findings

There are 340 files involved in exact duplication (7,013 unique MD5 values), none of them across folders, which agrees with the dataset description. Near-duplicates, however, are common: we find 3,484 strict pairs, 1,270 of them across the Training/Testing boundary, and 5,962 linked pairs (2,158 across). 441 Testing images (27.6%) have a strict near-duplicate in Training and 620 (38.8%) are linked to one. The overlap is class-dependent: strict cross-split duplicates affect 327 no-tumour, 105 meningioma, 9 glioma and 0 pituitary Testing images. 13 images belong to groups that carry more than one class label. Figure 1 shows examples. In addition, 100 Training and 103 Testing meningioma files are augmented copies (Section 5.1).

3.4 Leakage-aware split

We remove from the training pool every Training image whose group contains a Testing image (1,198 images) and every image in a label-conflict group, then assign whole groups to validation (about 15% per class, seeded) and the rest to training. This leaves 3,739 training and 663 validation images, and keeps the official 1,600 Testing images untouched, so the test set is directly comparable with other work while no linked image of it is seen in training. The removal is very uneven (Figure 2): 836 of the 1,400 no-tumour Training images are linked to Testing images, leaving only 478 for training. We also report results on the Testing folder reduced to one image per group (1,347 images).

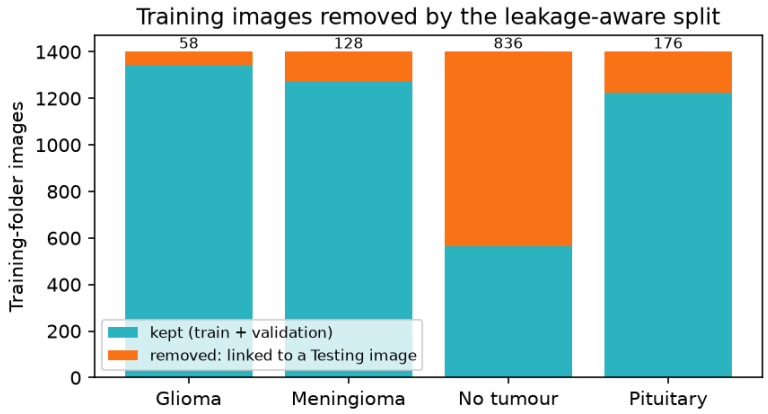


Figure 2: Training-folder images kept and removed by the leakage-aware split, per class.

4 Method

Pre-processing. Every image is converted to greyscale (ITU-R 601 luma), zero-padded to a square to keep its aspect ratio, resized to 224×224 with bilinear filtering, replicated to three channels and normalised with ImageNet statistics. The browser demo implements the same steps.

Model and training. We fine-tune EfficientNet-B0 [25] pretrained on ImageNet (`timm` [26], `efficientnet_b0.ra_in1k`, 4.0 M parameters, dropout 0.2) in PyTorch [17] with AdamW (learning rate 5×10^{-4}, weight decay 10^{-4}), a one-cycle schedule, batch size 32 and 12 epochs. Inverse-frequency class weights compensate for the reduced no-tumour class. Augmentation consists of random resized crops (scale 0.8–1.0), horizontal flips, rotations up to 10° with 5% translation, and brightness/contrast jitter of 0.2. The checkpoint with the best validation macro-F1 is kept (epoch 12). All random generators are seeded (42). Training ran on an 8-core CPU without a GPU in about 125 s per epoch.

Naive comparison. To measure the effect of the split, we train an identical model on the original Training folder with a random, image-level 15% validation split and evaluate both models on the same Testing images.

Calibration and uncertainty. We fit a single temperature on the validation logits by minimising the negative log-likelihood [10]. A prediction is flagged *uncertain* when its calibrated confidence is below a threshold τ, chosen on validation as the smallest value in $\{0.50, 0.51, \ldots, 0.99\}$ for which the retained validation predictions reach at least 99% accuracy. For the leakage-aware model $\tau = 0.67$ (validation coverage 98.5%, retained accuracy 99.1%). The normalised entropy of the calibrated distribution is reported alongside but not thresholded.

Metrics. We report accuracy and macro-F1 with 95% bootstrap confidence intervals (1,000 resamples), per-class precision, recall and F1, the confusion matrix, one-vs-rest ROC-AUC and the expected calibration error (ECE, 15 bins). All numbers in this paper are generated by scripts from a single saved prediction file per model (`results/predictions/`); the same file feeds the demo's statistics.

5 Results

5.1 Classification

Table 1 summarises accuracy on the Testing folder and its subsets, and Table 2 gives per-class results for the leakage-aware model; the confusion matrix and ROC curves are in Figure 3.

Table 1: Accuracy, macro-F1 (95% bootstrap CIs) and calibrated ECE on the Testing folder and subsets. "Has strict dup. in Training" = Testing images with a strict near-duplicate in the original Training folder; "No linked dup." = Testing images with no linked image in it. Macro-F1 is omitted for the class-skewed subsets.

Model	Test subset	n	Accuracy [95% CI]	Macro-F1 [95% CI]	ECE
Leakage-aware	Full Testing	1600	0.934 [0.921, 0.945]	0.933 [0.921, 0.944]	0.036
Leakage-aware	Testing, one per group	1347	0.931 [0.918, 0.945]	0.926 [0.912, 0.940]	0.038
Leakage-aware	Has strict dup. in Training	441	0.925 [0.900, 0.952]	– (class-skewed subset)	0.044
Leakage-aware	No linked dup. in Training	980	0.927 [0.908, 0.943]	– (class-skewed subset)	0.043
Naive	Full Testing	1600	0.961 [0.951, 0.970]	0.961 [0.950, 0.970]	0.022
Naive	Testing, one per group	1347	0.959 [0.948, 0.970]	0.959 [0.947, 0.970]	0.023
Naive	Has strict dup. in Training	441	0.995 [0.989, 1.000]	– (class-skewed subset)	0.005
Naive	No linked dup. in Training	980	0.941 [0.926, 0.953]	– (class-skewed subset)	0.035

Table 2: Per-class results of the leakage-aware model on the full Testing folder (n = 1,600).

Class	Precision	Recall	F1	ROC-AUC
Glioma	0.997	0.840	0.912	0.974
Meningioma	0.937	0.897	0.917	0.988
No tumour	0.857	1.000	0.923	0.996
Pituitary	0.966	0.998	0.982	1.000
Macro average	–	–	0.933	0.989

The leakage-aware model reaches 93.4% accuracy and macro-F1 0.933 on the full Testing folder, and 93.1% on the one-per-group variant. Errors are not symmetric. Pituitary tumours are almost always recognised (recall 99.8%), and every no-tumour image is classified correctly (recall 100.0%), but at a cost in no-tumour precision (85.7%): 39 gliomas and 28 meningiomas are called "no tumour". Glioma recall is the lowest (84.0%) although its precision is 99.7%. Missing a tumour is the clinically more serious error, and in this model it is the most frequent one. Validation accuracy (98.6%) is much higher than test accuracy, which points to a shift between the Training and Testing folders beyond the duplicates we removed (for instance different source collections or acquisition protocols); we could not resolve the source of each image.

Naive versus leakage-aware training. The model trained on the original Training folder scores 96.1% on the same Testing folder (95% CI 95.1%–97.0%), 2.8 points (44 images) above the leakage-aware model. Two effects are mixed in this gap. The naive model sees more data (4,760 instead of 3,739 training images, and 1,190 instead of 478 no-tumour images), which can genuinely help. But Table 3 shows where the gap comes from: on the 105 Testing meningiomas that have a strict near-duplicate in Training, the naive model is right on 105/105 and the leakage-aware model on 74/105, whereas on the meningiomas without one the two are close (288/295 versus 285/295). Most of the headline difference is therefore recall of near-copies of training images rather than better generalisation. No-tumour images are classified correctly by both models whether or not they have a duplicate, so the large no-tumour overlap does not change accuracy here, although it would affect any comparison that relies on that class.

Augmented files. The meningioma class contains files named `*-aug-*` (100 in Training, 103 in Testing), which appear to be augmented copies. Searching all eight flips and 90° rotations, only 26 of them match another image under our "linked" rule (10 only after a flip or rotation), so most are transformed in ways our hash does not detect. The leakage-aware model is correct on 99.0% of the augmented Testing meningiomas but only 86.5% of the others, which suggests that some leakage through augmented variants remains in our split.

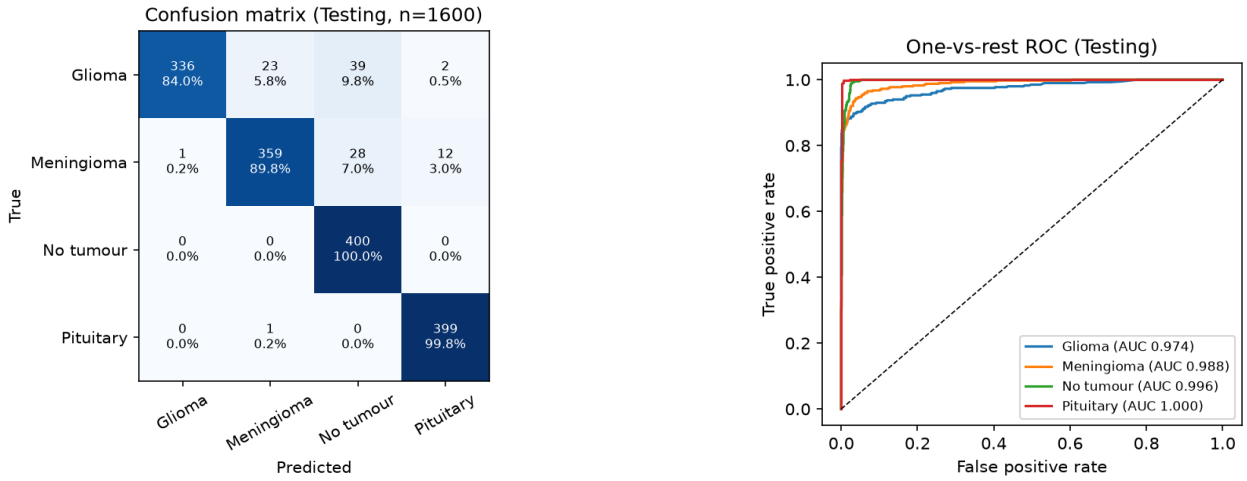


Figure 3: Leakage-aware model on the Testing folder: confusion matrix (counts and row percentages) and one-vs-rest ROC curves.

Table 3: Accuracy (correct/total) per class on Testing images with and without a strict near-duplicate in the original Training folder.

	Strict dup. in Training		No strict dup.	
Class	Leakage-aware	Naive	Leakage-aware	Naive
Glioma	0.778 (7/9)	0.778 (7/9)	0.841 (329/391)	0.867 (339/391)
Meningioma	0.705 (74/105)	1.000 (105/105)	0.966 (285/295)	0.976 (288/295)
No tumour	1.000 (327/327)	1.000 (327/327)	1.000 (73/73)	1.000 (73/73)
Pituitary	–	–	0.998 (399/400)	0.998 (399/400)

5.2 Calibration and uncertainty

The fitted temperature is 1.84, meaning the raw model is over-confident. On Testing, ECE falls from 0.054 to 0.036 (Figure 4). With $\tau = 0.67$, 3.8% of Testing predictions are flagged uncertain; accuracy is 95.3% on the confident ones and 44.3% on the flagged ones. The flag therefore concentrates errors, but most errors on Testing still carry high confidence, so it cannot be treated as a safety net.

6 Explainability

6.1 Methods

All explanations target the predicted class. **Grad-CAM** weights the final $7 \times 7 \times 1280$ feature map by the spatially averaged gradients; because EfficientNet-B0 ends in global average pooling and a linear layer, these weights equal the classifier weights divided by 49, so Grad-CAM coincides with CAM up to a positive scale. We use this to compute Grad-CAM in ONNX without gradients (maximum difference from autograd Grad-CAM after normalisation 7.4e-06). **Grad-CAM++** uses the closed-form higher-order weights of Chattopadhay et al. [6]. **Integrated Gradients** (Captum [12]) integrates 32 steps from a black image. **Occlusion** slides a black 32×32 patch with stride 16 and records the drop in the class logit. **KernelSHAP** [14] attributes the class logit to about 64 SLIC superpixels [1] using 200 coalitions with a black baseline. Signed maps are clipped (occlusion, SHAP) or taken in absolute value (IG), then min–max normalised.

6.2 Evaluation protocol

On 100 Testing images (25 per class, seeded) we compute the *deletion* and *insertion* AUC [18]: pixels are removed (set to black) or revealed (from a Gaussian-blurred copy) in 32 equal steps in

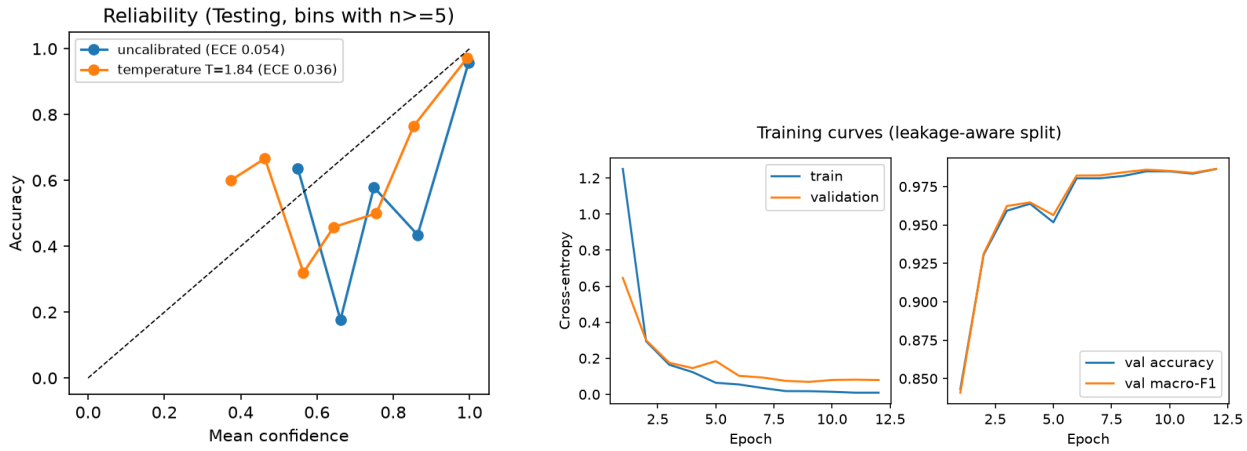


Figure 4: Left: reliability diagram before and after temperature scaling (Testing). Right: training curves of the leakage-aware model.

Table 4: Faithfulness (mean ± s.d. over 100 Testing images), attribution mass on the head region, and Spearman correlation with the trained-model map after randomising the classifier only and after randomising the whole network (12 images).

Method	Deletion AUC ↓	Insertion AUC ↑	Mass on head region	ρ after classifier rand.	ρ after full rand.
Grad-CAM	0.442 ± 0.359	0.828 ± 0.215	76.3%	0.26	-0.08
Grad-CAM++	0.454 ± 0.346	0.823 ± 0.217	81.8%	0.91	0.05
Integrated Gradients	0.260 ± 0.213	0.680 ± 0.259	97.7%	0.94	0.84
Occlusion	0.387 ± 0.385	0.790 ± 0.254	83.3%	0.15	0.02
KernelSHAP (SLIC)	0.419 ± 0.369	0.764 ± 0.276	73.4%	0.17	0.17
Random baseline	0.285 ± 0.106	0.424 ± 0.342	57.9%	–	–

order of decreasing attribution, and the softmax probability of the target class is integrated over the fraction of pixels. A random attribution serves as baseline. We also measure the share of attribution mass on the bright head region (a crude foreground mask). For the sanity check [2] we cumulatively re-initialise the network from the classifier down to the stem on 12 images and compute the Spearman rank correlation between each method's map from the trained and the randomised model; a method that passes should lose correlation as layers are randomised.

6.3 Results

Table 4 and Figure 5 summarise the comparison.

Insertion separates the methods clearly from the random baseline (0.424): Grad-CAM (0.828) and Grad-CAM++ (0.823) rank the pixels that restore the prediction best, followed by occlusion (0.790), KernelSHAP (0.764) and Integrated Gradients (0.680). *Deletion* does not: only Integrated Gradients (0.260) beats the random baseline (0.285), and the region-based methods score worse than random. With a black fill, scattered random deletions create out-of-distribution images that the model abandons quickly, while deleting a compact region leaves a natural-looking slice; the deletion metric therefore rewards pixel-scattered maps under this baseline and should be read with care. Integrated Gradients puts 97.7% of its mass on the head region, the other methods 73–83%.

The sanity check is more decisive. Randomising only the classifier already drops the correlation with the trained-model map to 0.26 for Grad-CAM, 0.15 for occlusion and 0.17 for KernelSHAP, so these maps depend on what the model learned. Grad-CAM++ stays at 0.91 after the classifier is randomised and only falls to 0.05 once the whole network is random, meaning that its maps are largely determined by the feature extractor rather than the class decision. Integrated Gradients fails the test: its maps still correlate at 0.84 with the original after every layer has been re-initialised, the edge-detector behaviour described by Adebayo et al. [2]. Its good deletion score is thus not evidence that it reflects the model. Agreement between methods is moderate at best (Spearman

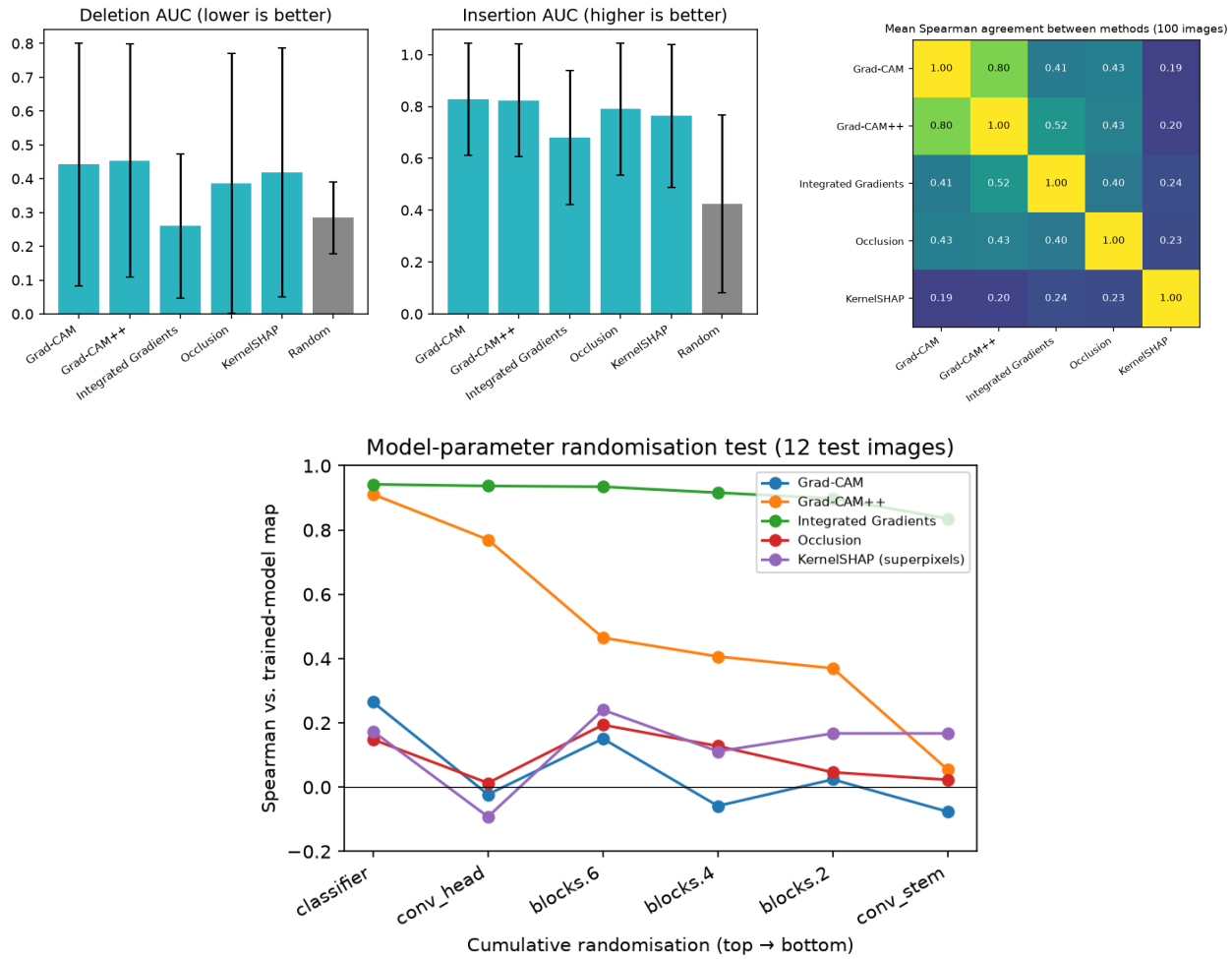


Figure 5: Top: deletion/insertion AUC per method and mean pairwise Spearman agreement. Bottom: cascading model-parameter randomisation.

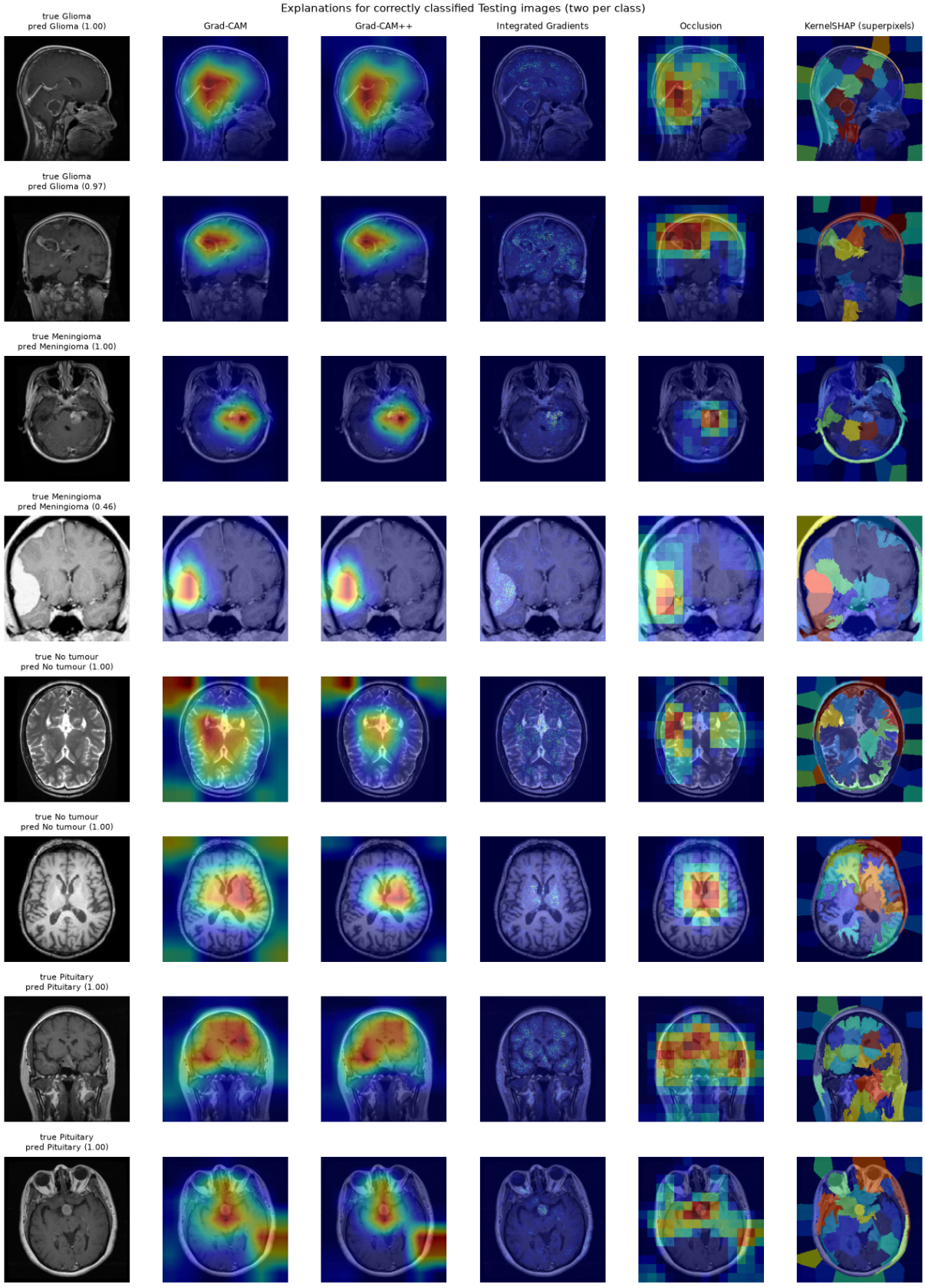
0.80 between Grad-CAM and Grad-CAM++, 0.43 with occlusion, 0.41 with Integrated Gradients and 0.19 with KernelSHAP).

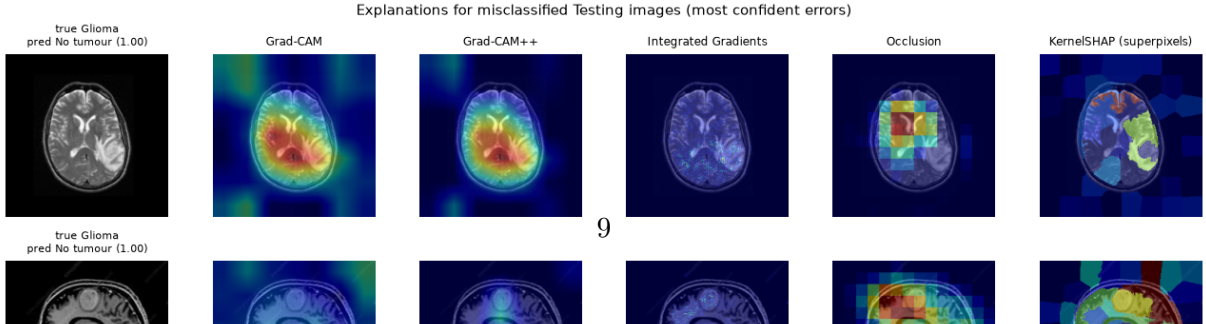
Qualitatively (Figure 6), for correct predictions Grad-CAM and occlusion usually highlight the lesion or the sellar region for pituitary tumours. Several confident errors are informative: in gliomas called "no tumour" the maps concentrate on the ventricles or outside the brain rather than on the visible lesion, consistent with the model missing it. We therefore use Grad-CAM, which passes the sanity check and leads on insertion, as the default in the demo, offer occlusion as a model-agnostic cross-check, and present the other maps only as precomputed comparisons.

7 In-browser demonstrator

The leakage-aware model is exported to ONNX (16.0 MB) with two outputs, the logits and the four class activation maps, and runs in the browser with ONNX Runtime Web (WebAssembly); the maximum logit difference to PyTorch on 64 Testing images is 3.5e-04. The page replicates the pre-processing, applies the fitted temperature and shows animated class probabilities, a Grad-CAM overlay with an opacity control and an on-demand occlusion map (a 7×7 grid of 32-pixel patches, 49 forward passes). Eight Testing slices (two per class, drawn from images with no linked image in Training) are bundled with their true labels and with precomputed Grad-CAM++, Integrated Gradients and KernelSHAP maps.

A *Why this prediction* panel is generated from templates filled only with measured quantities: the predicted class and calibrated confidence, the margin over the runner-up class, the uncertainty


Explanations for correctly classified Testing images (two per class)
true Glioma
pred Glioma (1.00)
Grad-CAM
Grad-CAM++
Integrated Gradients
Occlusion
KernelSHAP (superpixels)
true Glioma
pred Glioma (0.97)
true Meningioma
pred Meningioma (1.00)
true Meningioma
pred Meningioma (0.46)
true No tumour
pred No tumour (1.00)
true No tumour
pred No tumour (1.00)
true Pituitary
pred Pituitary (1.00)
true Pituitary
pred Pituitary (1.00)


Explanations for misclassified Testing images (most confident errors)
true Glioma
pred No tumour (1.00)
Grad-CAM
Grad-CAM++
Integrated Gradients
Occlusion
KernelSHAP (superpixels)
true Glioma
pred No tumour (1.00)

flag ($\tau = 0.67$) and normalised entropy, the share of Grad-CAM mass on the head region and its dominant image quadrant, the rank agreement between methods for the bundled samples, and that class's test precision and recall with its most common confusion, read from the same statistics file as Table 2. It contains no medical reasoning and is not a diagnosis.

8 Limitations

- **No patient-level split.** The dataset has no patient identifiers. Our groups capture near-identical images and very similar neighbouring slices, but other slices of the same patient may still occur on both sides of the split, so results may remain optimistic.
- **Augmented copies.** Augmented meningioma files in both folders are mostly not detected by our hash and probably still leak information (Section 5.1).
- **Thresholds by inspection.** The duplicate thresholds were set by looking at examples. Different thresholds change the counts; the released candidate pairs allow re-analysis.
- **Source and protocol confounding.** No-tumour images come from a different collection than the tumour classes, so a model can partly learn source characteristics (contrast, resolution, framing) rather than pathology. The gap between validation and test accuracy is consistent with such shifts.
- **Slice-level, single-label task.** Real diagnosis uses volumes, several sequences, clinical context and histopathology; this model sees one 2-D slice and four labels and cannot recognise other conditions.
- **Single run.** We trained one seed per configuration on a CPU; the confidence intervals reflect test-set sampling only, not training variance.
- **Explanations are not localisation.** No tumour masks were used, so we cannot measure whether attributions fall on the lesion; the head-region share is only a sanity proxy.

9 Ethical considerations

The images are public and de-identified, and we redistribute only eight resized samples under the CC BY 4.0 licence with attribution. A tumour classifier can cause harm if its output is mistaken for a diagnosis: it can miss tumours (the most common error here) and give confident wrong answers. The demonstrator therefore states prominently that it is a research and education tool, not a medical device, processes images locally without uploading them, shows calibrated uncertainty, and builds its rationale only from the model's own outputs and measured error rates. Saliency maps can create unwarranted trust [2, 3]; we report their sanity checks and faithfulness rather than presenting them as evidence of correctness.

10 Conclusion

A widely used brain MRI benchmark that reports no train/test overlap at the file level still shares near-duplicate images between its folders, mostly in the no-tumour class. With those removed from training and the training data split by duplicate group, a CPU-trained EfficientNet-B0 reaches 93.4% accuracy on the official Testing folder, with its errors concentrated in tumours called "no tumour". Calibration, an explicit uncertainty threshold, tested explanations and a rationale built from measured statistics make the behaviour of the model easier to inspect, but they do not make it a clinical tool. Patient-level splits, external test sets and lesion annotations are the next requirements for any stronger claim.

Code and data availability. Code, notebooks, split files, hashes, predictions and the demo: https://github.com/Rishabh-bgp/brain-tumor-mri-classification. Live demo: https://rishabh-bgp.github.io/brain-tumor-mri-classification/. Dataset: Nickparvar [16].

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